

Parametric Survival Modeling

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents parametric survival models fitted for the METIMMOX-1 economic evaluation. The models compare different biomarker strategies for overall survival (OS) and progression-free survival (PFS) outcomes. This analysis serves as the foundation for the cost-effectiveness evaluation by quantifying treatment effects across biomarker-defined patient subgroups.

2 Population Characteristics including Biomarker Prevalences

Population characteristics not available. Run main analysis first.

3 Model Selection and Comparison

3.1 Analysis Specifications

- **Distributions tested:**
 1. Exponential
 2. Weibull
 3. Weibull PH
 4. Gompertz
 5. Log-logistic
 6. Log-normal
 7. Generalized Gamma
 8. Gamma
 9. Generalized F Distribution
- **Covariate structure:** Full models with age and sex adjustment, treatment effects, and treatment-biomarker interactions:
 1. Age (continuous, years)
 2. Sex (categorical, 0=female, 1=male)
 3. Treatment assignment (Control arm vs Experimental arm)
 4. CRP biomarker status (0=negative, 1=positive)
 5. TLR biomarker status (0=negative, 1=positive)
 6. TMB_BRAF biomarker status (0=negative, 1=positive)
 7. Treatment $\times$ CRP interaction
 8. Treatment $\times$ TLR interaction
 9. Treatment $\times$ TMB_BRAF interaction

Note: Models without age and sex adjustment were also considered but are not used in the primary analysis.

3.2 Data Considerations

- **Complete case analysis:** Only patients with complete covariate data included
- **Reference patient:** Modal categories for categorical variables, mean for continuous variables
- **Time horizon:** Models fitted to observed follow-up period with extrapolation capability
- **Biomarker prevalences:** Taken from the METIMMOX-1 trial and may or may not reflect real world settings

3.3 Model Validation and Goodness of Fit

- **Selection criterion:** AIC is more appropriate here because:

- BIC penalizes model complexity more heavily (uses $\log(n)$ vs 2 for AIC)
- For prediction purposes, we want the model that best balances fit and complexity
- BIC tends to select overly simple models that might miss important biomarker interactions
- AIC is better for predictive modeling while BIC is better for identifying the “true” model
- **Validation:** Visual comparison with Kaplan-Meier curves and clinical plausibility assessment

4 Model Comparison

Table 1: Overall Survival (OS) model comparison using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	622.56	646.47	0.00	1.0	***
Full Model	weibull	623.44	647.36	0.88	2.5	
Full Model	weibullph	623.44	647.36	0.88	2.5	
Full Model	llogis	623.68	647.60	1.12	4.0	
Full Model	lognormal	624.09	648.01	1.53	5.0	
Full Model	gengamma	624.56	650.65	2.00	6.0	
Full Model	genf	626.56	654.82	4.00	7.0	
Full Model	gompertz	633.09	657.01	10.53	8.0	
Full Model	exponential	654.14	675.88	31.58	9.0	

Table 2: Progression-Free Survival (PFS) model comparison using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	453.43	477.35	0.00	1.0	***
Full Model	lognormal	453.46	477.38	0.03	2.0	
Full Model	weibullph	454.19	478.11	0.76	3.5	
Full Model	weibull	454.19	478.11	0.76	3.5	
Full Model	llogis	455.16	479.08	1.73	5.0	
Full Model	gengamma	455.28	481.37	1.85	6.0	
Full Model	genf	457.28	485.55	3.85	7.0	
Full Model	gompertz	467.57	491.49	14.14	8.0	
Full Model	exponential	483.52	505.26	30.09	9.0	

5 Model Coefficients

5.1 Overall Survival (OS) Model Coefficients

Model comparison:

- Best-fitting distribution: gamma
- Covariate adjustment: Age and sex adjusted

Parameter	Gamma			Weibull PH		
	Estimate	SE	exp(Est)	Estimate	SE	exp(Est)
Age (per year)	0.0015	0.0082	1.0015	0.0105	0.0168	1.0106
rate	-2.8975	0.6157	0.0552	NA	NA	NA
RxControl	-0.3570	0.3162	0.6998	-0.8624	0.6443	0.4222
arm:crp1						
RxControl	-0.8039	0.2509	0.4476	-1.6010	0.4921	0.2017
arm:tlr1						
RxControl	-0.0866	0.2210	0.9171	-0.1394	0.4303	0.8699
arm:tmb_braf1						
RxExperimental	-0.2309	0.2852	0.7939	-0.4313	0.5275	0.6497
arm						
RxExperimental	-0.3395	0.2265	0.7122	-0.6574	0.4639	0.5182
arm:crp1						
RxExperimental	-0.4109	0.2129	0.6630	-0.7773	0.4178	0.4596
arm:tlr1						
RxExperimental	-0.1859	0.2072	0.8303	-0.2591	0.4186	0.7717
arm:tmb_braf1						
scale	NA	NA	NA	-9.3901	1.5795	0.0001
Sex (male vs female)	0.1454	0.1482	1.1565	0.3426	0.2988	1.4086
shape	1.2275	0.1820	3.4126	0.7281	0.1090	2.0711

5.2 Progression-Free Survival (PFS) Model Coefficients

Model comparison:

- Best-fitting distribution: gamma
- Covariate adjustment: Age and sex adjusted

Parameter	Gamma			Weibull PH		
	Estimate	SE	exp(Est)	Estimate	SE	exp(Est)
Age (per year)	-0.0009	0.0084	0.9991	-0.0032	0.0169	0.9968
rate	-1.8647	0.6321	0.1549	NA	NA	NA
RxControl	0.0654	0.4253	1.0676	0.4137	0.8152	1.5123
arm:crp1						
RxControl	-0.9764	0.2718	0.3767	-2.0111	0.5566	0.1338
arm:tlr1						
RxControl	-0.1055	0.2537	0.8998	-0.2783	0.4940	0.7571
arm:tmb_braf1						
RxExperimental	0.4218	0.3073	1.5247	0.9743	0.5936	2.6492
arm						
RxExperimental	-0.5754	0.2412	0.5625	-1.0969	0.5515	0.3339
arm:crp1						
RxExperimental	-1.1914	0.2133	0.3038	-2.3671	0.4623	0.0938
arm:tlr1						
RxExperimental	-0.4891	0.2325	0.6132	-1.2349	0.5657	0.2909
arm:tmb_braf1						
scale	NA	NA	NA	-6.7666	1.3964	0.0012
Sex (male vs female)	0.1321	0.1653	1.1413	0.2264	0.3463	1.2541
shape	1.2426	0.1901	3.4647	0.7437	0.1139	2.1038

6 Population Survival Curves

The following figures compare empirical Kaplan-Meier estimates with predicted survival curves from the selected parametric models. “Population” refers to different patient groups: biomarker-positive patients, biomarker-negative patients, or the entire population. For the control strategy, population curves represent all patients. For biomarker-guided strategies, population-level predictions are weighted by the prevalence of biomarker-positive and biomarker-negative patients in the sample.

6.1 Model Validation: Comparison with Kaplan-Meier Estimates

These figures show how well each tested parametric distribution fits the observed data for the entire study population.

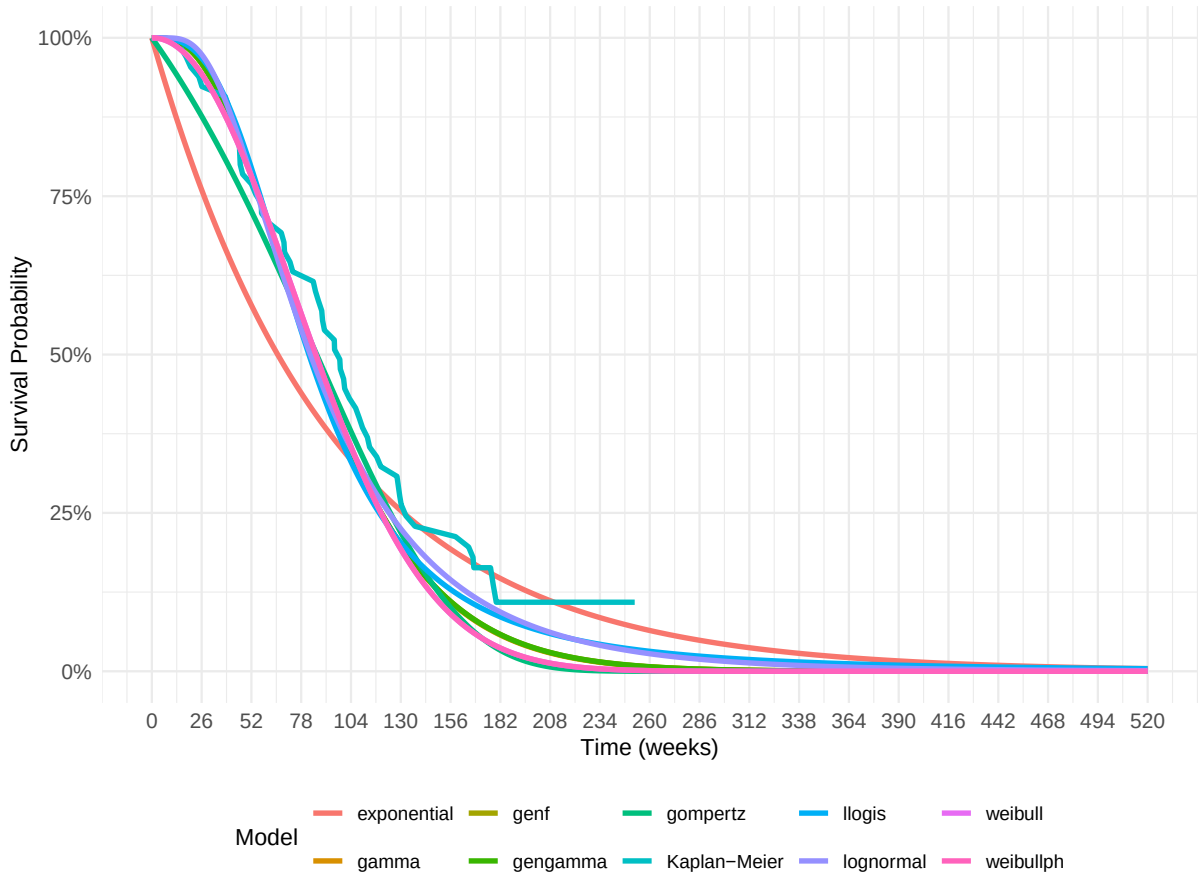


Figure 1: Overall survival curves for age- and sex-adjusted models compared to Kaplan-Meier estimates. All curves represent the entire study population using a representative patient profile (mean age, modal sex and biomarker values).

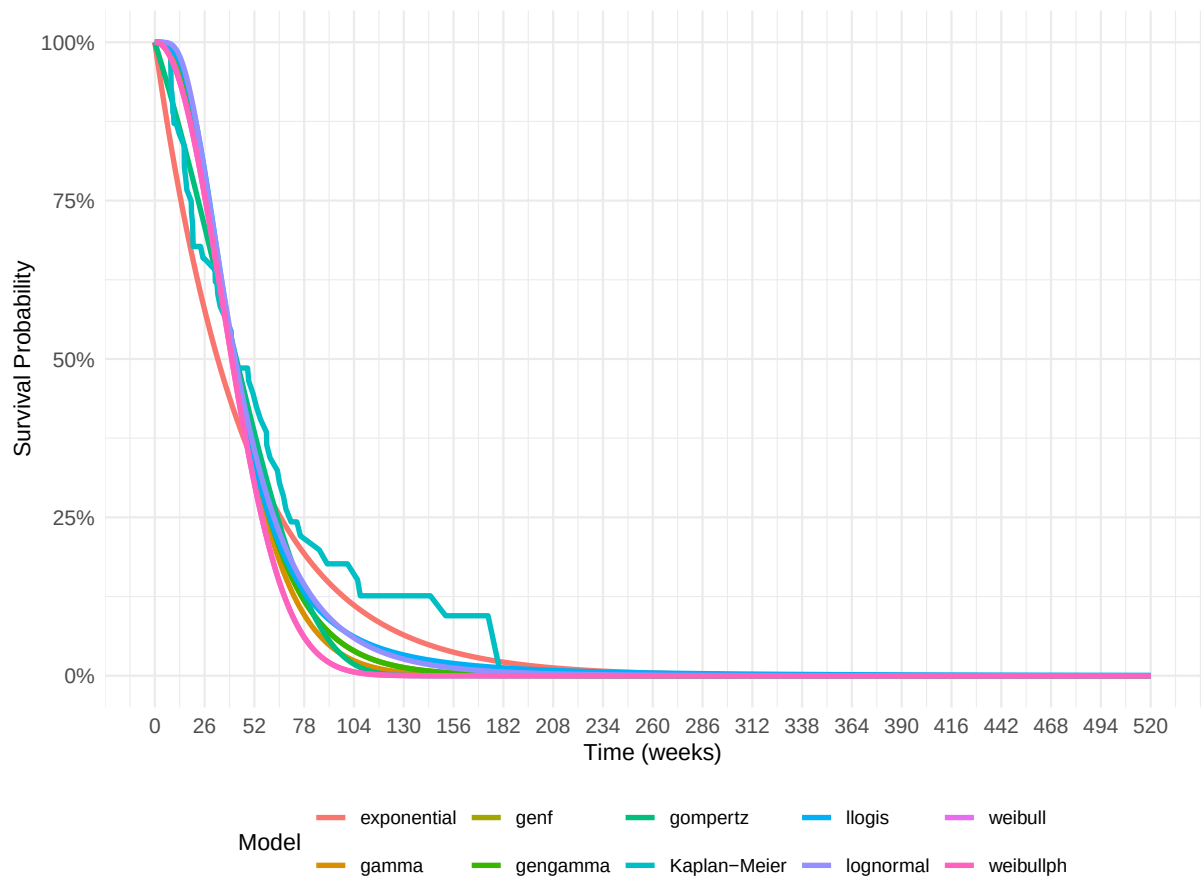


Figure 2: Progression-free survival curves for age- and sex-adjusted models compared to Kaplan-Meier estimates. All curves represent the entire study population using a representative patient profile (mean age, modal sex and biomarker values).

7 Biomarker-Specific Results

The following figures compare survival outcomes between biomarker-positive and biomarker-negative patient subgroups for each biomarker strategy. These comparisons inform the potential value of biomarker-guided treatment selection. Kaplan-Meier estimates are shown for biomarker-positive patients receiving experimental treatment and biomarker-negative patients receiving control treatment, alongside parametric model predictions.

7.1 CRP Biomarker Strategy

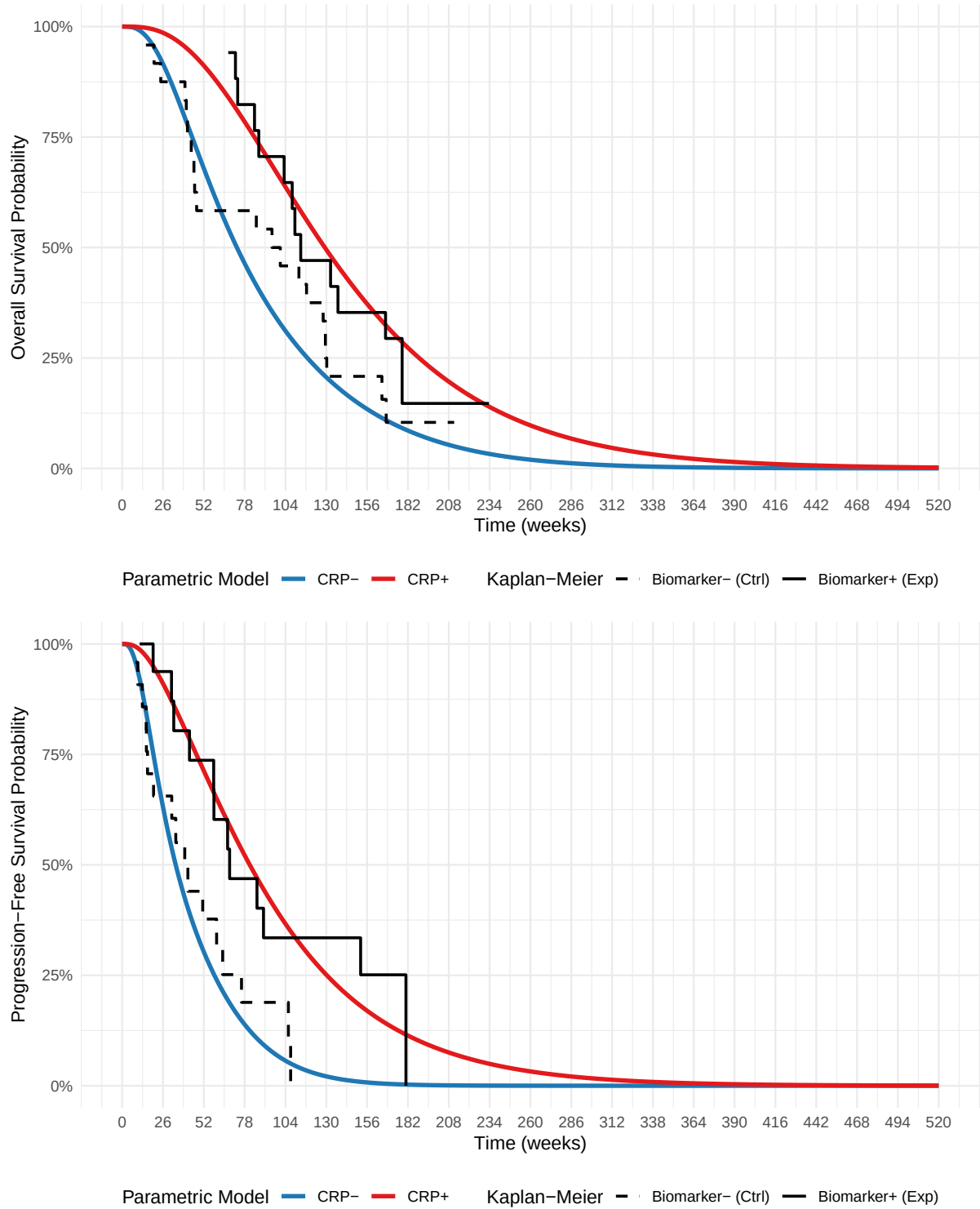


Figure 3: Overall survival and progression-free survival for CRP biomarker-positive versus biomarker-negative patients. Colored curves show predicted survival from the selected parametric model. Black curves show Kaplan-Meier estimates for the treated (biomarker-positive, experimental arm, solid line) and untreated (biomarker-negative, control arm, dashed line) subgroups. Differences between curves indicate the predictive value of CRP status for treatment benefit.

7.2 TLR Biomarker Strategy

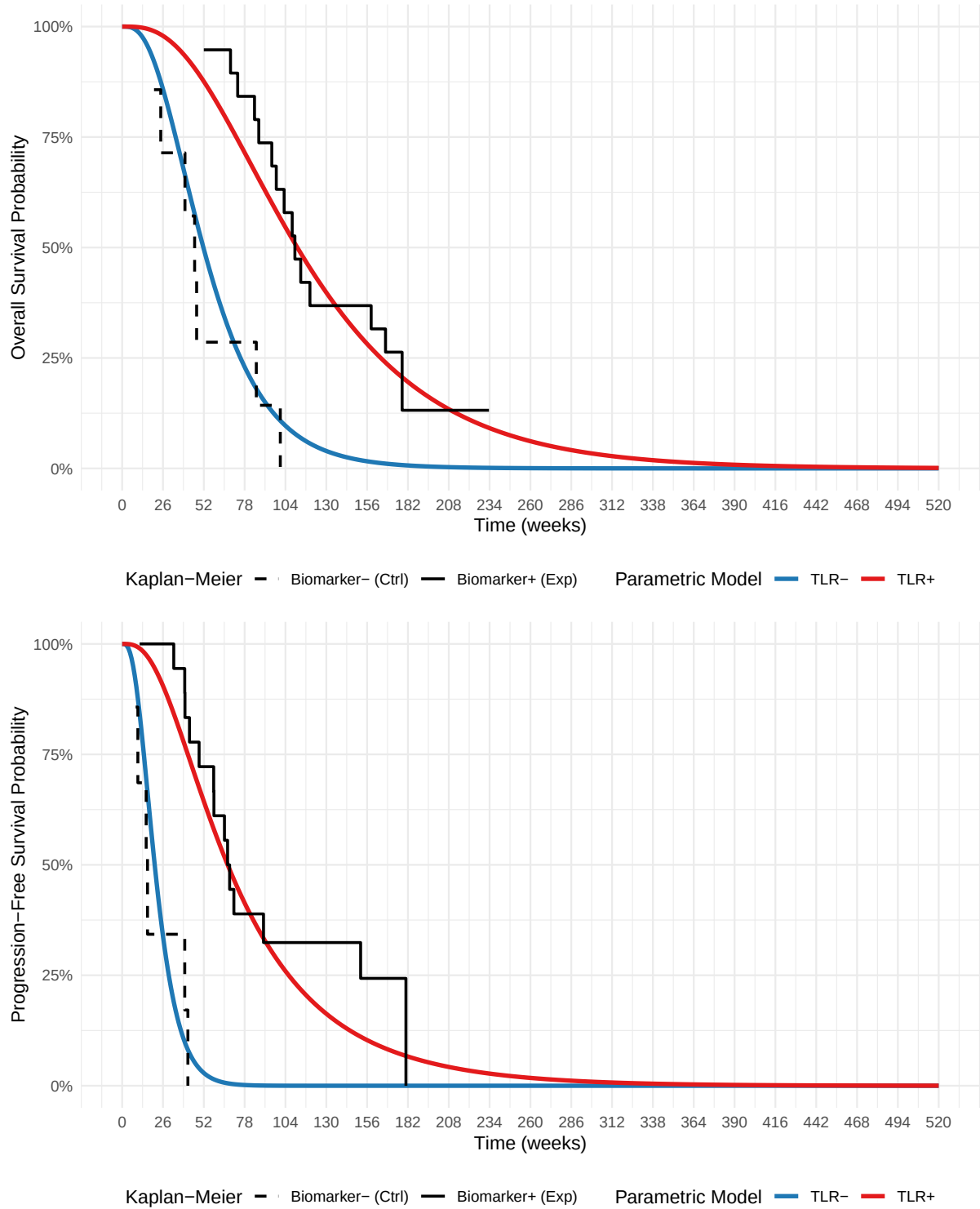


Figure 4: Overall survival and progression-free survival for TLR (target lesion reduction) biomarker-positive versus biomarker-negative patients. Colored curves show predicted survival from the selected parametric model. Black curves show Kaplan-Meier estimates for the treated (biomarker-positive, experimental arm, solid line) and untreated (biomarker-negative, control arm, dashed line) subgroups. Differences between curves indicate the predictive value of TLR status for treatment benefit.

7.3 TMB_BRAF Biomarker Strategy

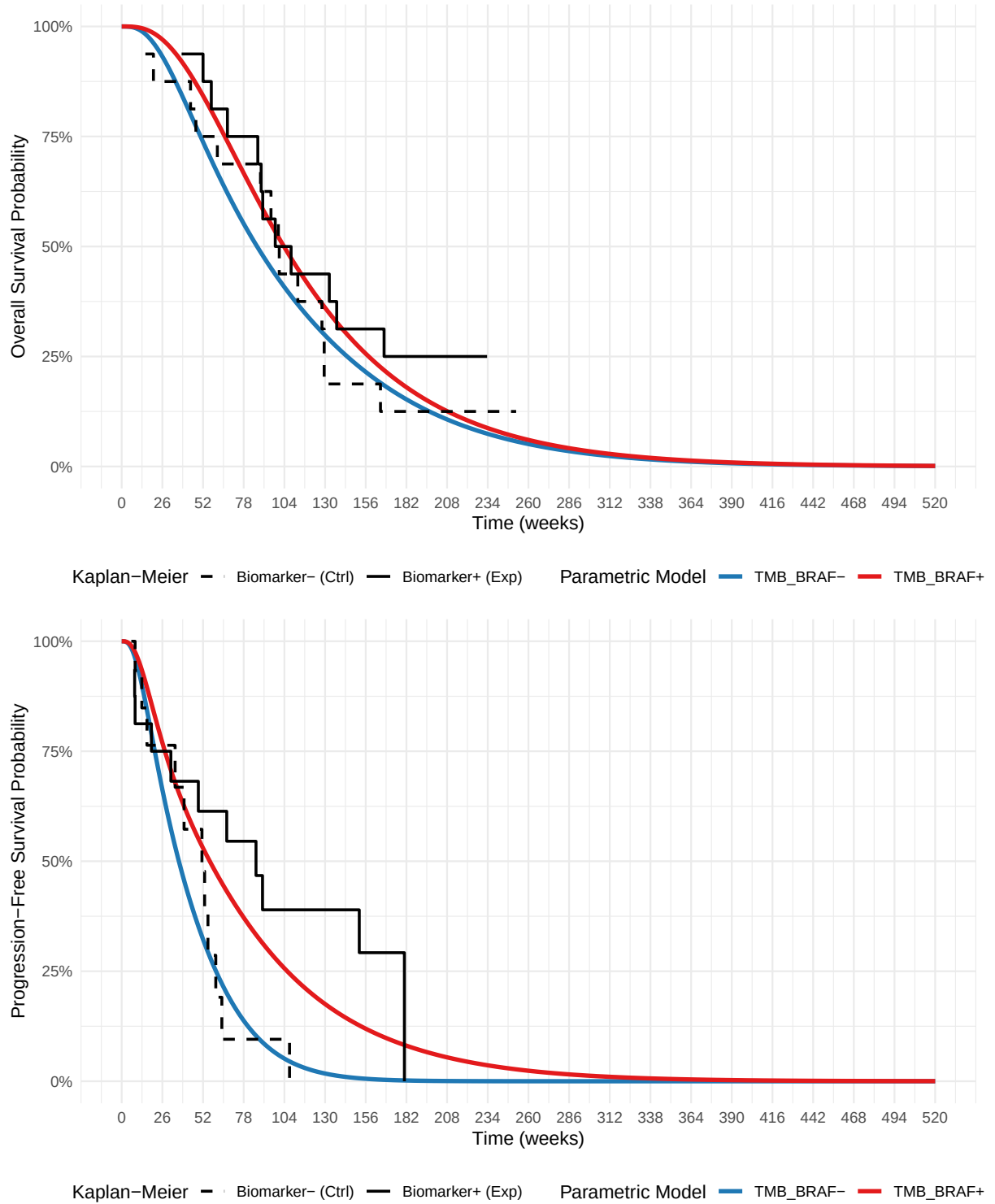


Figure 5: Overall survival and progression-free survival for TMB_BRAF biomarker-positive versus biomarker-negative patients. Colored curves show predicted survival from the selected parametric model. Black curves show Kaplan-Meier estimates for the treated (biomarker-positive, experimental arm, solid line) and untreated (biomarker-negative, control arm, dashed line) subgroups. Differences between curves indicate the predictive value of TMB_BRAF status for treatment benefit.

8 Strategy Comparison

The following figures compare all three biomarker-guided treatment strategies against the control (standard of care) strategy. These population-level predictions are weighted by biomarker prevalence to reflect expected outcomes in an unselected population.

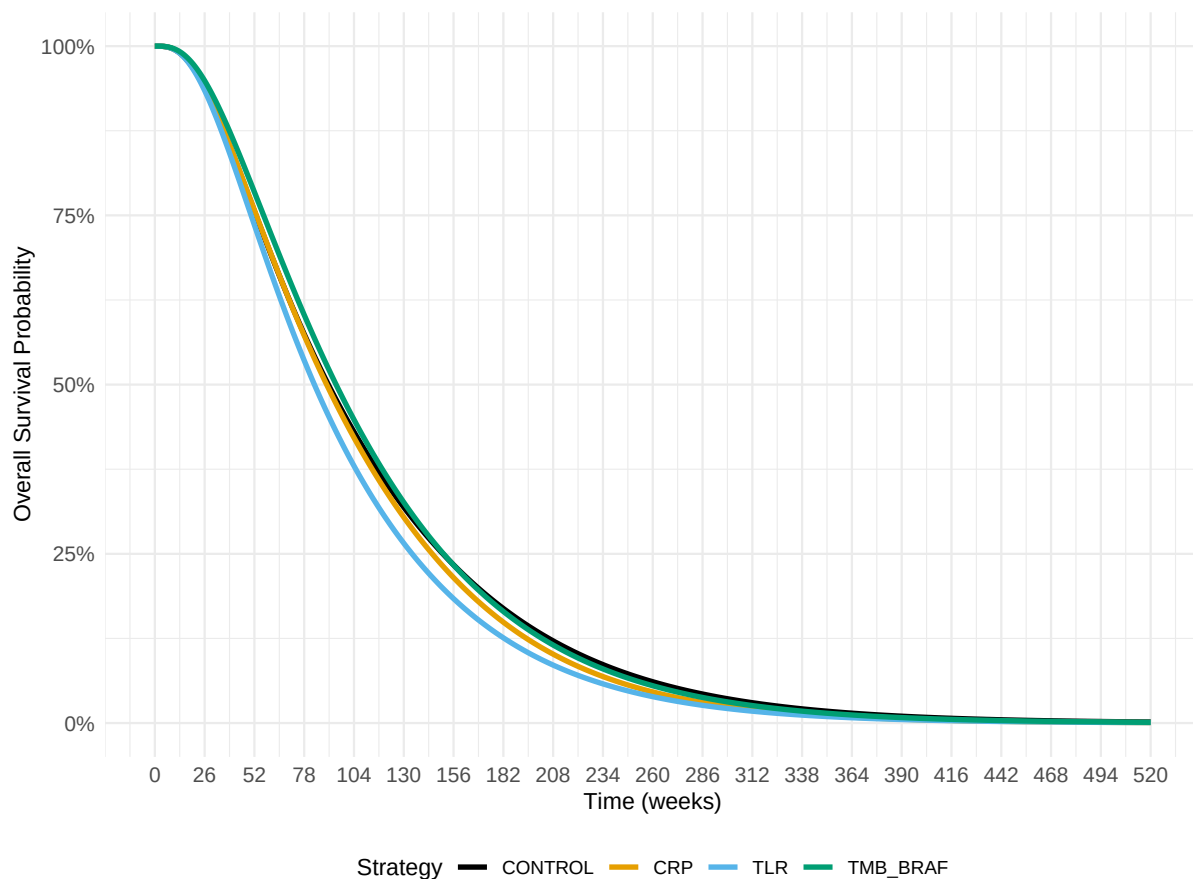


Figure 6: Overall survival comparison across all treatment strategies. Each curve represents the population-weighted expected OS for each strategy: Control (standard of care for all patients), CRP-guided treatment allocation, TLR-guided treatment allocation, and TMB_BRAF-guided treatment allocation. Differences between strategies reflect both the predictive accuracy of each biomarker and its prevalence in the population.

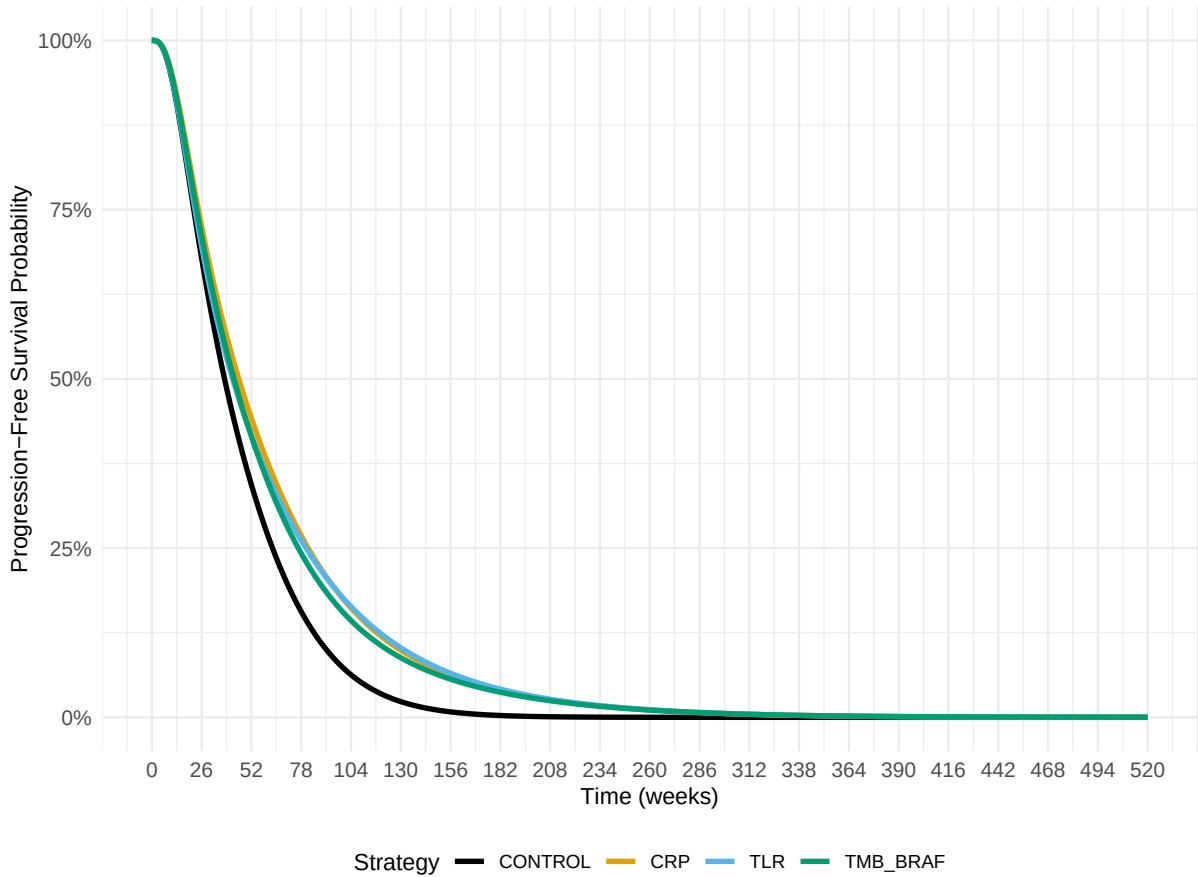


Figure 7: Progression-free survival comparison across all treatment strategies. Each curve represents the population-weighted expected PFS for each strategy: Control (standard of care for all patients), CRP-guided treatment allocation, TLR-guided treatment allocation, and TMB_BRAF-guided treatment allocation. Differences between strategies reflect both the predictive accuracy of each biomarker and its prevalence in the population.

8.1 Unadjusted Life-Years by Strategy

The following table shows unadjusted life-years (area under the survival curve) for each treatment strategy. Population-weighted values reflect the expected outcomes accounting for biomarker prevalence.

Table 3: Unadjusted Life-Years by Treatment Strategy (Population-Weighted)

Strategy	OS Life-Years	PFS Life-Years	OS Gain vs Control	PFS Gain vs Control
CONTROL	2.16	0.89	0.00	0.00
CRP	2.08	1.19	-0.08	0.30
TLR	1.96	1.16	-0.20	0.27
TMB_BRAF	2.19	1.13	0.03	0.24

Note: Life-years are calculated as the area under the survival curve (restricted mean survival time) over the model time horizon. These are unadjusted values without utility weighting. Population-weighted values account for biomarker prevalence in each strategy.

9 Sensitivity Analysis: Separate Biomarker Models

9.1 Rationale

The **all-in-one model** includes all biomarkers and their treatment interaction terms simultaneously:

$$\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{crp:Rx} + \text{tlr:Rx} + \text{tmb_braf:Rx}$$

When predicting survival for a biomarker-positive patient (e.g., CRP+), the model also activates OTHER biomarker main effects and interaction terms if that patient happens to be positive for those biomarkers. Since biomarkers are correlated (e.g., approximately 65% of patients are TLR-positive), this can attenuate or amplify the estimated treatment effect for a specific biomarker strategy.

This sensitivity analysis fits **separate biomarker-specific models** where each model only includes the biomarker of interest and its treatment interaction:

- CRP model: $\sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{crp:Rx}$
- TLR model: $\sim \text{Age} + \text{sex} + \text{Rx} + \text{tlr} + \text{tlr:Rx}$
- TMB_BRAF model: $\sim \text{Age} + \text{sex} + \text{Rx} + \text{tmb_braf} + \text{tmb_braf:Rx}$

All models use the **gamma distribution**, selected by AIC in the primary analysis.

9.2 Hazard Ratio Comparison

The following table compares the treatment $\times$ biomarker interaction hazard ratios between the all-in-one model and separate models.

Interpretation: $\text{HR} < 1$ indicates reduced hazard (improved survival) with experimental treatment in biomarker-positive patients. $\text{HR} = 1.0$ is the reference (no interaction effect), meaning biomarker-positive patients have the same treatment effect as biomarker-negative patients receiving control treatment.

Table 4: Treatment $\times$ Biomarker Interaction Hazard Ratios: All-in-One vs Separate Models

Biomarker	Outcome	All-in-One	Separate
CRP	OS	0.700 (0.377-1.301)	0.577 (0.300-1.107)
CRP	PFS	1.068 (0.464-2.457)	0.834 (0.297-2.342)
TLR	OS	0.448 (0.274-0.732)	0.420 (0.257-0.686)
TLR	PFS	0.377 (0.221-0.642)	0.393 (0.216-0.715)
TMB_BRAF	OS	0.917 (0.595-1.414)	1.043 (0.640-1.699)
TMB_BRAF	PFS	0.900 (0.547-1.479)	0.857 (0.432-1.700)

9.3 Biomarker-Positive vs. Biomarker-Negative Subgroups

The following figures compare survival predictions between biomarker-positive and biomarker-negative subgroups for each biomarker strategy, using both the all-in-one model and separate models. Control strategy and Kaplan-Meier estimates are shown for reference.

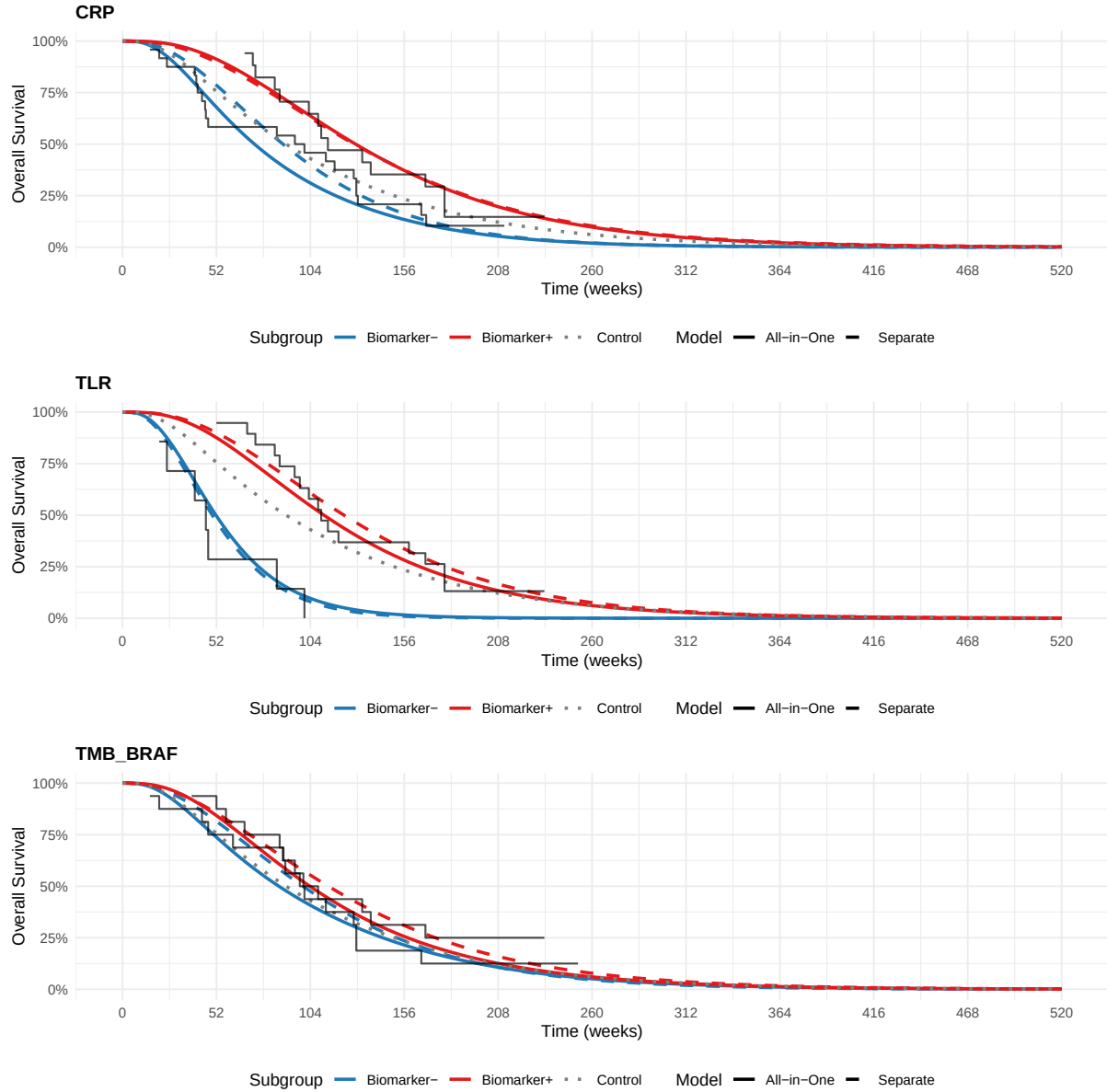


Figure 8: Overall survival: Biomarker-positive vs. biomarker-negative vs. control. Solid lines: All-in-One model. Dashed lines: Separate model. Red: Biomarker-positive. Blue: Biomarker-negative. Gray dotted: Control. Black step lines: Kaplan-Meier estimates.

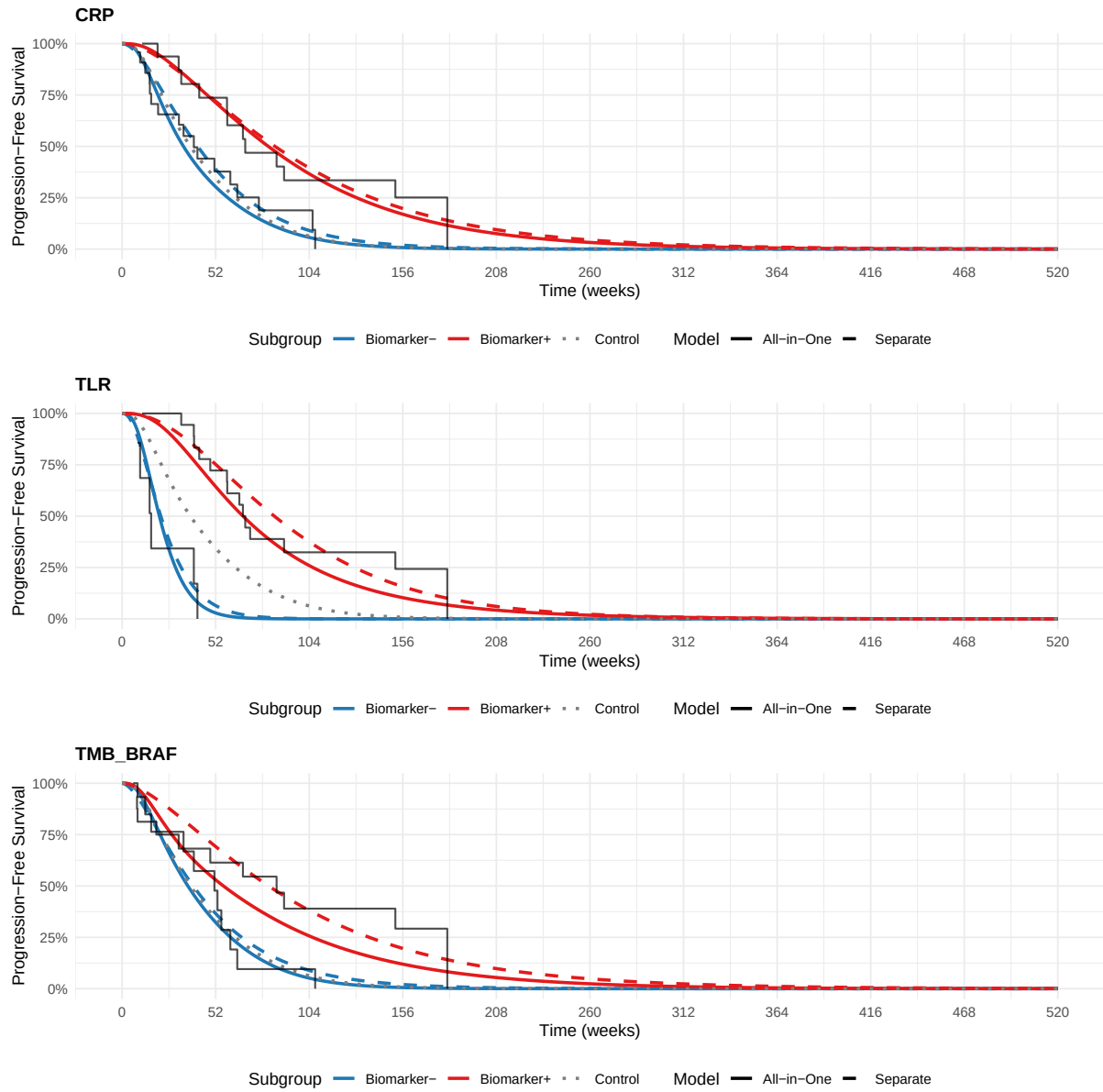


Figure 9: Progression-free survival: Biomarker-positive vs. biomarker-negative vs. control. Solid lines: All-in-One model. Dashed lines: Separate model. Red: Biomarker-positive. Blue: Biomarker-negative. Gray dotted: Control. Black step lines: Kaplan-Meier estimates.

9.4 Strategy-Level Comparison: All-in-One vs Separate Models

The following figures compare population-weighted survival predictions between the all-in-one model and separate models for each biomarker strategy, with the control strategy as reference.

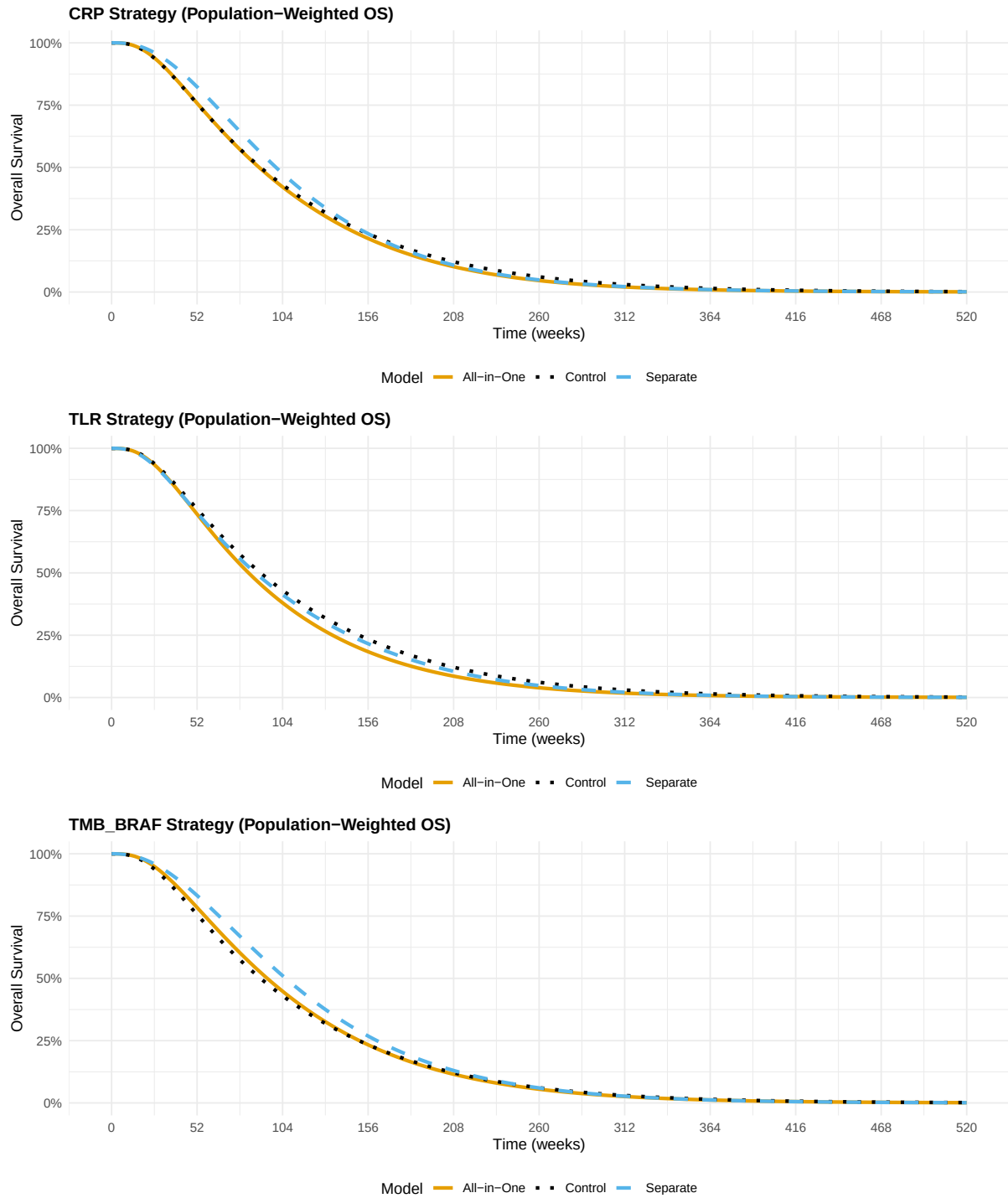


Figure 10: Population-weighted overall survival by biomarker strategy comparing All-in-One model, Separate model, and Control strategy. Each panel shows one biomarker strategy with population-weighted predictions that account for biomarker prevalence.

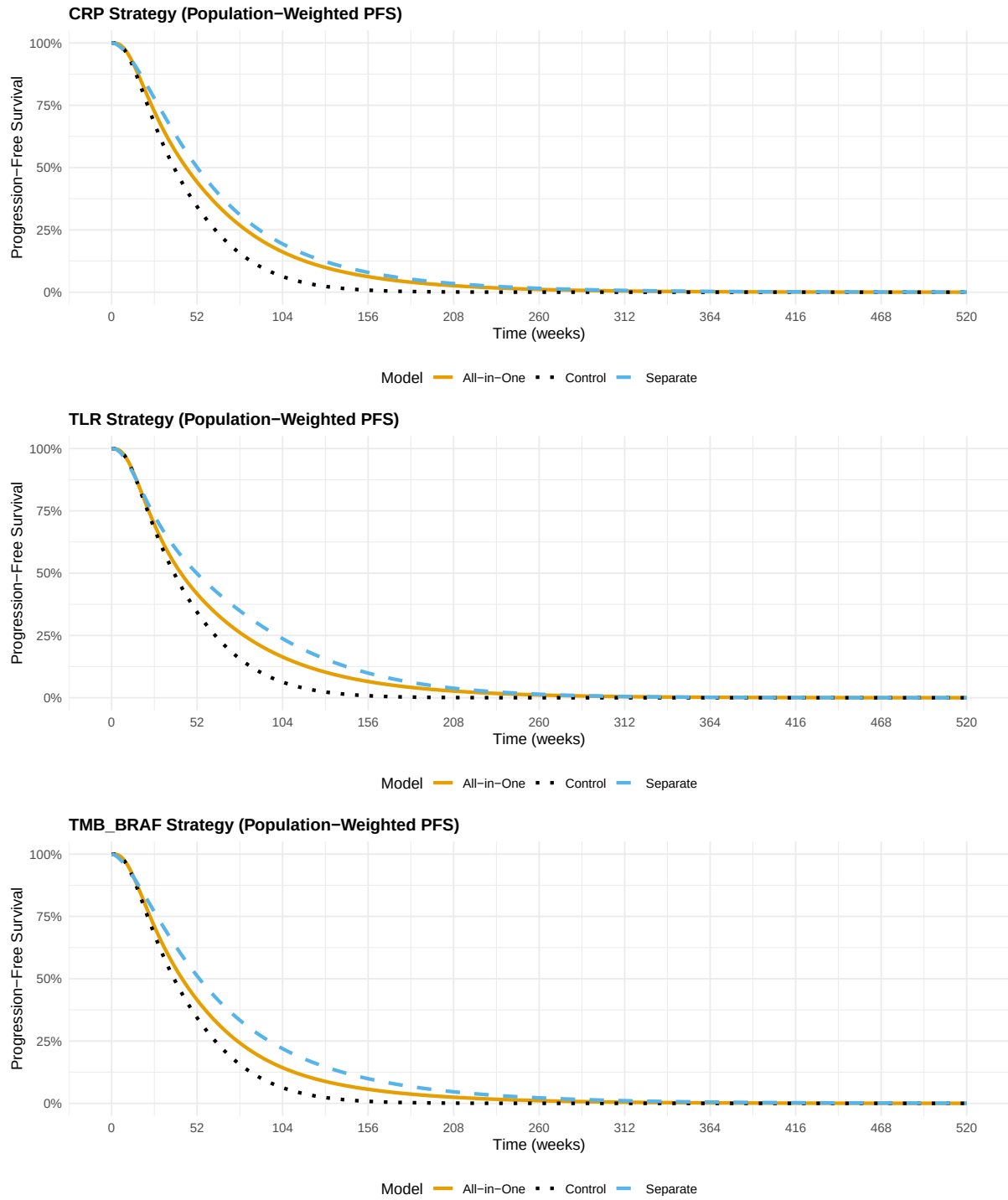


Figure 11: Population-weighted progression-free survival by biomarker strategy comparing All-in-One model, Separate model, and Control strategy. Each panel shows one biomarker strategy with population-weighted predictions that account for biomarker prevalence.

9.5 Mean Survival Comparison

The following tables show mean survival (restricted mean survival time, i.e., area under the survival curve) comparing the all-in-one model versus separate models for each biomarker strategy. These are unadjusted life-years without utility weighting.

9.5.1 Biomarker Subgroups: Overall Survival

Table 5: Mean Overall Survival (Unadjusted Life-Years) by Subgroup

Subgroup	All-in-One	Separate	Difference	% Change
Control	2.164	2.164	NA	NA
CRP+	2.800	2.797	-0.003	-0.1
CRP-	1.716	1.940	0.224	13.1
TLR+	2.450	2.642	0.192	7.8
TLR-	1.125	1.072	-0.053	-4.7
TMB_BRAF+	2.338	2.523	0.185	7.9
TMB_BRAF-	2.066	2.213	0.147	7.1

9.5.2 Biomarker Subgroups: Progression-Free Survival

Table 6: Mean Progression-Free Survival by Subgroup

Subgroup	All-in-One	Separate	Difference	% Change
Control	0.891	0.891	NA	NA
CRP+	1.881	1.984	0.103	5.5
CRP-	0.835	0.980	0.145	17.4
TLR+	1.587	1.882	0.295	18.6
TLR-	0.442	0.479	0.037	8.4
TMB_BRAF+	1.483	1.950	0.467	31.5
TMB_BRAF-	0.853	0.946	0.093	10.9

9.5.3 Overall Strategy Comparison

The following tables compare population-weighted outcomes across all strategies, sorted by incremental gain vs control. Population-weighted values account for biomarker prevalence in each strategy. The control strategy is model-independent (interaction terms do not activate when all patients receive control treatment).

Table 7: Overall Survival (Population-Weighted)

Strategy	All-in-One	Separate	vs Control	
	Life-Years	—	Incr. (A-i-O)	Incr. (Sep)
Control	2.164	NA	0.000	NA
TLR	1.961	2.063	-0.203	-0.101
CRP	2.082	2.230	-0.082	0.066
TMB_BRAF	2.188	2.351	0.024	0.187

Table 8: Progression-Free Survival (Population-Weighted)

Strategy	All-in-One	Separate	vs Control	
	PFS Years	—	Incr. (A-i-O)	Incr. (Sep)
Control	0.891	NA	0.000	NA
TMB_BRAF	1.134	1.394	0.243	0.503
TLR	1.164	1.364	0.273	0.473
CRP	1.189	1.320	0.298	0.429

10 Summary

- Parametric survival models were fitted for both overall survival (OS) and progression-free survival (PFS) outcomes
- Model selection was based on Akaike Information Criterion (AIC) to optimize predictive performance
- Age- and sex-adjusted models with treatment-biomarker interactions were used in the primary analysis
- Biomarker-specific survival patterns were characterized for CRP, TLR (target lesion reduction), and TMB_BRAF strategies
- Visual validation against Kaplan-Meier estimates confirms good model fit across all tested distributions
- Population-level predictions account for biomarker prevalence to reflect expected outcomes in clinical practice
- Strategy comparison shows relative performance of biomarker-guided treatment allocation approaches
- Sensitivity analysis comparing all-in-one vs separate biomarker models assesses robustness to biomarker correlation effects

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